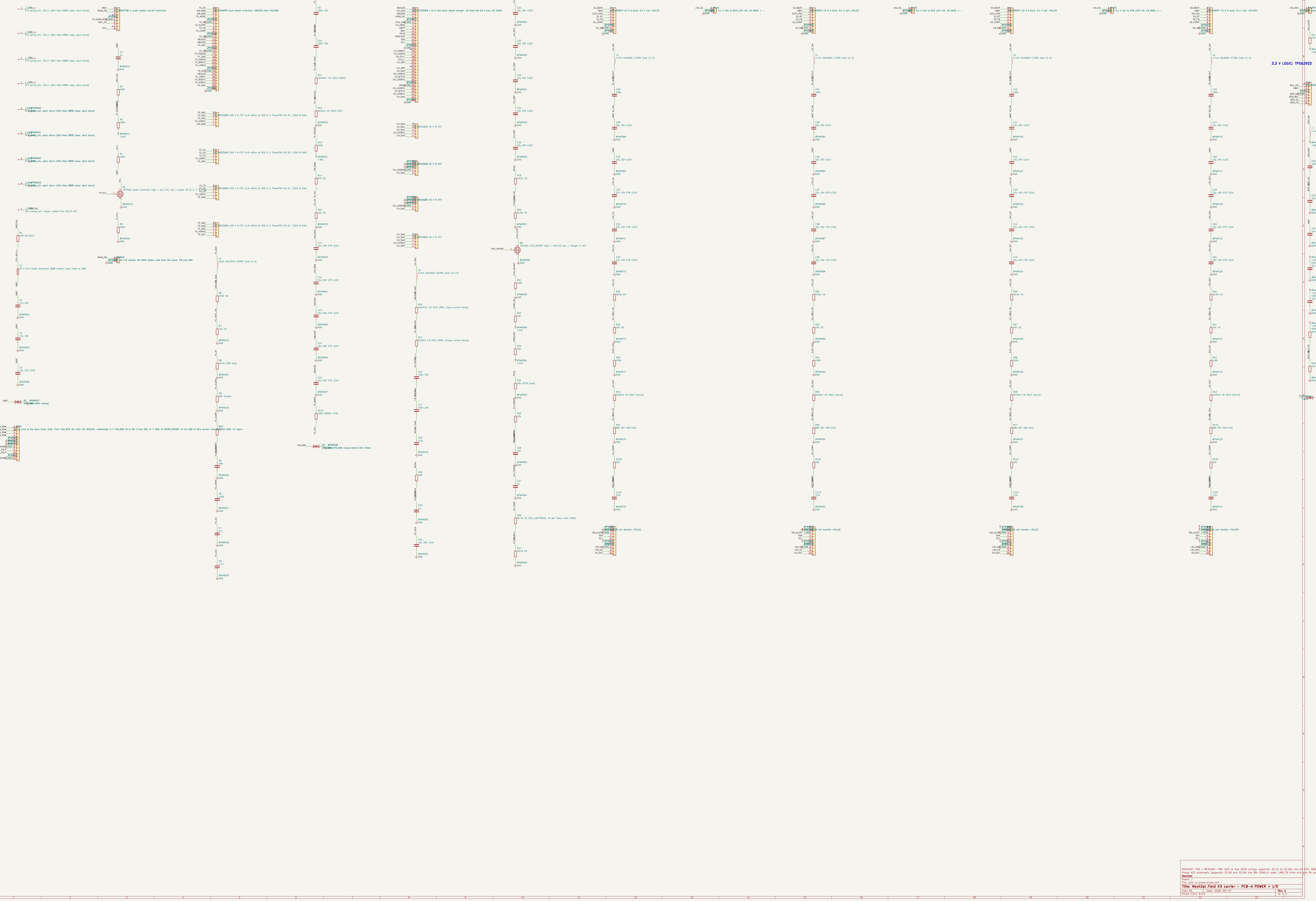

PACK NODE OVER THE DOCK BLOCK (32.56): 9 A PINS, PRE–CHARGE, 25 A BLADE FUSE
PI–POWER CONTROL: LTC2954 SIGNAL PINS
FRONT END: LM5176 FROM THE 9 TO 36 V INPUT TO THE 20 V CHARGE BUS
CHARGER BQ25731: 4S FROM THE 20 V BUS INTO THE PACK NODE, I2C 0x6B
SLOT RAIL S1: AP64500 5.1 V + INA226 0x40
SLOT RAIL S2: AP64500 5.1 V + INA226 0x41
SLOT RAIL S3: AP64500 5.1 V + INA226 0x44
DEVICE RAIL: AP64500 5.1 V + INA226 0x45
3.3 V LOGIC: TPS62933
J_CP1 CELL+ 9 A spring pin, CELL+ (Mill–Max 0858 class, dock block)
J_CP2 CELL+ 9 A spring pin, CELL+ (Mill–Max 0858 class, dock block)
J_CP3 CELL+ 9 A spring pin, CELL+ (Mill–Max 0858 class, dock block)
J_CP4 CELL+ 9 A spring pin, CELL+ (Mill–Max 0858 class, dock block)
J_CN1 9 A spring pin, pack return (Mill–Max 0858 class, dock block)
J_CN2 9 A spring pin, pack return (Mill–Max 0858 class, dock block)
J_CN3 9 A spring pin, pack return (Mill–Max 0858 class, dock block)
J_CN4 9 A spring pin, pack return (Mill–Max 0858 class, dock block)
J_PRE PRECHG pre–charge pin, longer, mates first (32.24 AX)
R1 10R 2W 2512
F1 25 A mini blade (Keystone 3568 holder): pack node to VBAT
C1 47u 25V
C2 47u 25V
C3 10u 25V 1210
D1 SMCJ26A (VBAT clamp)
J_DOCK spring pins to the dock block (2x6, Preci–Dip 813–S1–012–10–016101, underside): 1–4 VIN_RAW (9 to 36 V from E6), 5–7 GND, 8 SHORE_INHIBIT, 9–10 USB of E6's sensor controller, 11 GND, 12 spare
VIN_RAW
SHORE_INHIBIT
USB_E6_P
USB_E6_N
DOCK_SPARE
U1 LTC2954CTS8–1 push–button on/off controller
VBAT
MAIN_PB
PI_SHDN_REQ
RAIL_EN
KILL
C4 1u
R2 100k
R3 100k
R4 100k
Q1 2N7002: panel controller high = pull KILL low = power off (1 G, 2 S, 3 D)
PI_KILL
R5 100k
J_MAINSW JST–XH 1x2 socket: the MAIN button lead from the panel, PB and GND
U2 LM5176PWPR buck–boost controller, VBUS20 from VIN_RAW
FE_EN
FE_MODE
FE_RT
FE_SLOPE
FE_SS
FE_COMP
FE_FB
VBUS20
FE_OUT
FE_CS
FE_PGOOD
FE_SW2
FE_HDRV2
FE_BOOT2
FE_LDRV2
FE_VCC
FE_LDRV1
FE_BOOT1
FE_HDRV1
FE_SW1
Q2 CSD19532Q5B 100 V N–FET (4.6 mOhm at VGS 6 V, PowerPAK SO–8 / SON–8 5x6)
Q3 CSD19532Q5B 100 V N–FET (4.6 mOhm at VGS 6 V, PowerPAK SO–8 / SON–8 5x6)
Q4 CSD19532Q5B 100 V N–FET (4.6 mOhm at VGS 6 V, PowerPAK SO–8 / SON–8 5x6)
Q5 CSD19532Q5B 100 V N–FET (4.6 mOhm at VGS 6 V, PowerPAK SO–8 / SON–8 5x6)
L1 10uH XAL1010–103ME (Isat 14 A)
R6 240k 1%
R7 10k 1%
R8 40.2k (300 kHz)
R9 30k (slope)
R10 10k
C5 10n
C6 100p
C7 47n
C8 4.7u
C9 100n 25V
C10 100n 25V
R11 10mOhm 1% 2512 (ISNS)
R12 5mOhm 1% 2512 (CS)
R13 100k
R14 62k 1%
R15 10k 1%
C11 10u 50V X7R 1210
C12 10u 50V X7R 1210
C13 10u 50V X7R 1210
C14 10u 50V X7R 1210
C15 10u 50V X7R 1210
R119 100k (MODE: CCM)
D2 SMCJ33A (VIN_RAW clamp behind E6's filter)
U3 BQ25731RSNR 1 to 5 cell buck–boost charger, 4S from the 20 V bus, I2C 0x6B
CH_ACN
CHRG_OK
CHG_ILIM
CH_VDDA
IADPT
IBAT
PSYS
PROCHOT
SDA
SCL
CH_COMP1
CH_COMP2
CH_CELL
CELL+
CH_SRP
CH_SW2
CH_HIDRV2
CH_BTST2
CH_LODRV2
REGN
CH_LODRV1
CH_BTST1
CH_HIDRV1
CH_SW1
Q7 CSD18510Q5B 40 V N–FET
Q8 CSD18510Q5B 40 V N–FET
Q9 CSD18510Q5B 40 V N–FET
Q10 CSD18510Q5B 40 V N–FET
L2 3.3uH XAL6030–332ME (Isat 12.2 A)
R16 10mOhm 1% 2512 (RAC, input current sense)
R17 5mOhm 1% 2512 (RSR, charge current sense)
C16 100n 25V
C17 100n 25V
C18 3.3u
R18 10R
C19 1u
C20 10u 35V 1210
C21 10u 35V 1210
C22 10u 35V 1210
C23 22u 25V 1210
C24 22u 25V 1210
C25 22u 25V 1210
R19 16.5k 1%
R20 34.8k 1%
Q6 2N7002: CHG_INHIBIT high = ILIM_HIZ low = charger in HiZ
CHG_INHIBIT
R21 100k
R22 10k
R23 10k
R24 10k (PSYS load)
R25 10k
C26 10n
C27 1n
R26 60.4k 1% (CELL_BATPRESZ: 4S per Table, from VDDA)
R27 40.2k 1%
U4 AP64500SP–13 5 A buck, 5.1 V rail +5V_S1
S1_BOOT
SLOT_EN1
S1_RT
S1_FB
S1_COMP
S1_SW
J_5V_S1 5.1 V rail to B16 (JST–VH, 16 AWG): + –
L3 4.7uH XAL6060–472ME (Isat 11 A)
C28 100n
C29 10u 25V 1210
C30 10u 25V 1210
C31 22u 10V X7R 1210
C32 22u 10V X7R 1210
C33 22u 10V X7R 1210
R28 53.6k 1%
R29 10k 1%
R30 100k
R31 5mOhm 1% 2512 (shunt)
R45 68k (RT: 500 kHz)
R129 22k
C112 3.3n
U8 INA226 rail monitor +5V_S1
INA_ALERT
S1_OUT
U5 AP64500SP–13 5 A buck, 5.1 V rail +5V_S2
S2_BOOT
SLOT_EN2
S2_RT
S2_FB
S2_COMP
S2_SW
J_5V_S2 5.1 V rail to B16 (JST–VH, 16 AWG): + –
L4 4.7uH XAL6060–472ME (Isat 11 A)
C34 100n
C35 10u 25V 1210
C36 10u 25V 1210
C37 22u 10V X7R 1210
C38 22u 10V X7R 1210
C39 22u 10V X7R 1210
R32 53.6k 1%
R33 10k 1%
R34 100k
R35 5mOhm 1% 2512 (shunt)
R46 68k (RT: 500 kHz)
R130 22k
C113 3.3n
U9 INA226 rail monitor +5V_S2
S2_OUT
U6 AP64500SP–13 5 A buck, 5.1 V rail +5V_S3
S3_BOOT
SLOT_EN3
S3_RT
S3_FB
S3_COMP
S3_SW
J_5V_S3 5.1 V rail to B16 (JST–VH, 16 AWG): + –
L5 4.7uH XAL6060–472ME (Isat 11 A)
C40 100n
C41 10u 25V 1210
C42 10u 25V 1210
C43 22u 10V X7R 1210
C44 22u 10V X7R 1210
C45 22u 10V X7R 1210
R36 53.6k 1%
R37 10k 1%
R38 100k
R39 5mOhm 1% 2512 (shunt)
R47 68k (RT: 500 kHz)
R131 22k
C114 3.3n
U10 INA226 rail monitor +5V_S3
S3_OUT
U7 AP64500SP–13 5 A buck, 5.1 V rail +5V_DEV
SD_BOOT
DEV_EN
SD_RT
SD_FB
SD_COMP
SD_SW
L6 4.7uH XAL6060–472ME (Isat 11 A)
C46 100n
C47 10u 25V 1210
C48 10u 25V 1210
C49 22u 10V X7R 1210
C50 22u 10V X7R 1210
C51 22u 10V X7R 1210
R40 53.6k 1%
R41 10k 1%
R42 100k
R43 5mOhm 1% 2512 (shunt)
R115 68k (RT: 500 kHz)
R132 22k
C115 3.3n
U11 INA226 rail monitor +5V_DEV
SD_OUT
J_5V_DEV
R44 10k
U12 TPS62933
B33_SW
B33_BST
B33_SS
B33_FB
L7 4.7u
C52 100n
C53 10n
C54 10u
C55 22u
C56 22u
R48 31.6
R49 10k
+3V3
GND

MESHSAT–709 / MESHSAT–789. A19 (4 Sep 2026 rulings, appendix 32.13 to 32.25): the kit UPS. BQ2
Phase A22 schematic (appendix 32.55 and 32.56: the BB–2590/U node, LM5176 front end and PA an

MeshSat

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Title: MeshSat Field Kit carrier – PCB–A POWER + I/O

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